MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, FLEXIBLE, NONTWISTABLE,
CLASS 7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the latest issue of Specification MIL-W-287.

REQUIREMENTS: See table I.

Part number: See table I.

Extent of qualification: Qualification obtained for one assembly shall constitute qualification for other assemblies as specified in table II.

Substitution data: See table III.

FSC 5985

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TABLE I. Flexible, nonbristable waveguide assemblies.

Group number	1/ Part number M287/7	TE_{10} mode frequency range (GHz)	Minimum centerline bending radius (inch)		Insertion loss (dB/ft)	Pressurization (lb_f/in^2 gage	2/ V (ma: a'	
			E plane	H plane			12"	24"
1	01-XXXX	3.5-8.2	3.00	6.00	0.10	30	1.20	1.25
	02-XXXX	4.75-11.0	1.75	3.50	0.15	35	1.20	1.25
	03-XXXX	7.5-18.0	1.625	3.25	0.20	45	1.20	1.25

Flexure (degrees/ft) $1x10^5$ cycles		Jacket material	Retention force (lb_f)	Oper temp range -55 to	Mates with standard rigid waveguide M23351/4	Flanges	
E plane	H plane					Flange 1 M39000/( )	Flange 2 M39000/( )
30	15	Neoprene	N/A	+100°C	029	3-058 (gasket)	3-032 (cover)
30	15	Neoprene	N/A	+100°C	033	3-070 (gasket)	3-044 (cover)
30	15	Neoprene	N/A	+100°C	037	3-082 (gasket)	3-074 (cover)

1/ The complete part number shall be the number shown and a four-digit number indicating the nominal relaxed length in inches.
2/ The VSWR value for any length between the lengths specified shall be the next higher value (e.g. for a 15-inch length of M287/7-01, the VSWR would be 1.25:1).
3/ AL = aluminum alloy.

TABLE II. Grouping for qualification.

Group number	Qualifying dash number	Dash numbers of assemblies qualified
1	02	01 and 03

TABLE III. Substitution data 1/

Dash number	Manufacturer's code (FSCM)	Manufacturer's part number
01	87864	TDS-29005 series
02	87864	TDS-22107 series
03	87864	TDS-14112 series

1/ Assemblies covered by this specification sheet are substitutable for the manufacturer's part numbers shown. This information in no way implies that the manufacturer's part is substitutable for the military part.

SWR		Material 3/
36"	>36"	
1.30	1.38	AL
1.30	1.38	AL
1.30	1.38	AL

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Custodians:
Army - EP
Navy - EC
Air Force - 85

Review activities:
Army - AV, MI
Navy - OS
Air Force - 17, 99
DLA - ES

User activities:
Army - AR
Navy - AS, CG, MC
Air Force - 11, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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DOCUMENT IDENTIFIER AND TITLE MIL-W-287/7 - Waveguide Assemblies, Flexible, Nontwistable, Class 7

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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